

Abhijeet Anand, PhD

Independent Fellow, Inter-University Centre for Astronomy and Astrophysics (IUCAA), Pune

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Astrophysicist and Data Scientist with a PhD in Astrophysics and 6+ years of experience building scalable machine learning models, analytics workflows, and data pipelines for large, complex datasets, including 60M+ records and 5+ TB of data per day. Skilled in Python, SQL, statistical modeling, and predictive analytics, with a strong track record of improving model accuracy, reducing false positives, and delivering production-quality pipelines. Experienced in cross-functional collaboration, open-source software development, and communicating scientific results effectively to diverse audiences.

Work Experience

Independent Postdoctoral Fellow	<i>IUCAA, Pune</i>	<i>Dec 2025 - Present</i>
Affiliate Scientist	<i>Lawrence Berkeley National Lab, USA</i>	<i>Feb 2026 - Present</i>

- Working as an independent postdoctoral fellow, developing ML and data science methods for large scientific datasets and drafting scientific goals for upcoming national and international astronomical facilities.

Postdoctoral Fellow	<i>Lawrence Berkeley National Lab, USA</i>	<i>Sep 2022 - Present</i>
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Data Engineering, ML and Big Data

- Built and deployed scalable **large-scale predictive modeling pipeline** ([redrock](#), multi-class classification; PCA feature engineering) on 60M+ records, improving accuracy by 30% and cutting false positives by 30%.
- Designed controlled experiments to compare modeling approaches, analogous to A/B testing to guide data driven method selection. Also performed model evaluation metrics (ROC-AUC, precision-recall, etc).
- Designed **distributed real-time I/O pipelines (FITS, HDF5) to process 5+ TB/day, $\approx 100k$ rows/min** of structured and unstructured data, optimized for analytics in Python/pandas and SQL. Architecture is transferable to cloud-based systems (e.g., GCP, AWS).
- Developed and maintained **internal/external analysis libraries (open source)**, leading features, code reviews, and model improvements, and reporting workflows. Improved software quality via automated tests (unittest), documentation, and supporting reproducible analytics at scale.

Project Leadership and Scientific Impact

- Led **two cross-functional projects** with 15+ team members, coordinating software and deployment efforts to improve classification performance and reliability, enabling more accurate data driven decisions in a large scale production pipeline.
- Proposed and implemented data-driven improvements to production pipelines, securing a successful **2-year project extension**. Recognized as a top 10% contributor within the collaboration.
- Mentored junior analysts/researchers on experimental design, statistical validation, and production-grade code-bases.
- Communicated analytical findings to both technical and non-technical audiences to support data driven decisions within a large international scientific collaboration.

PhD Research Fellow	<i>Max Planck Institute for Astrophysics, Germany</i>	<i>Sep 2018 - Jul 2022</i>
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Data Engineering, ML and Big Data

- Built parallel **signal-processing models (matched-kernel detection)** for large sequential data, raising precision ($\geq 95\%$ purity) and reducing runtime from weeks to hours on 1M+ samples
- Developed and accelerated **non-linear regression pipelines (with Numba)** for parameter estimation, with rigorous model testing/validation and performance analysis.
- Ran large-scale statistical validation (hypothesis testing, bootstrapping) and experiment design to quantify uncertainty and build models.

Project Leadership and Scientific Impact

- Led **two large research projects**, from concept to publication, resulting in high-purity data products used by **50+ research teams worldwide**.
- Collaborated with computational physicists **to merge observational and simulated datasets**, enabling new insights into the physical origins of detected patterns.

Technical Skills

Programming	Python (NumPy, Pandas, SciPy, scikit-learn, Matplotlib), SQL, Version control collaboration (Git workflows), LaTeX, Jupyter, unittest
ML & Statistics	Predictive modeling (regression/classification), NMF/PCA/feature engineering, clustering; statistics & probability (hypothesis testing, bootstrapping, experiment design).
Data Engineering	Parallel I/O pipelines (FITS, HDF5), HPC (Slurm, NERSC), Cloud-ready workflow design, Automated schedulers (cron), debugging, dashboards, code optimization and documentation, Exploratory data analysis (EDA), visualization for insights and reporting
Open Source	Maintainer/contributor: qsoabsfind , redrock (27 GitHub stars), desispec (39 GitHub stars)
Soft Skills	Mentorship, Team leadership, Cross-team collaboration, Agile workflows

Education

PhD in Astrophysics	Max Planck Institute for Astrophysics, Garching, Germany	Sep 2018 - Jul 2022
BS - MS in Physics	Indian Institute of Science (IISc), Bangalore, India	Aug 2012 - Jun 2017